

Algorithms Problem Solving

Level 2

Digit Frequency

The background of the slide is a photograph of a modern building's interior. It features a large, curved glass ceiling that allows natural light to filter through, creating a pattern of light and shadow on the floor. In the foreground, the silhouettes of several people are visible, some standing and some sitting, suggesting a collaborative workspace or a public area. The overall tone is professional and tech-oriented.

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Definition

- ◆ Digit frequency means counting how many times a specific digit appears inside a number.
- ◆ For example, in `122342`, the digit `2` appears `3` times.

Explanation

- ◆ ``ReadNumber(string message)`` reads a number from the user and displays the message passed to it.
- ◆ The function uses a ``do-while`` loop to make sure the entered number is not negative.
- ◆ ``CountFrequency()`` checks the digits of the number one by one.
- ◆ ``N % 10`` extracts the last digit of ``N``.
- ◆ ``N = N / 10`` removes the last digit from ``N``.
- ◆ If ``Digit == Remainder``, the frequency counter ``freq`` is increased by ``1``.
- ◆ The loop continues until all digits have been checked.
- ◆ ``CountFrequency()`` returns the total number of occurrences.
- ◆ ``PrintResult()`` displays the digit, its frequency, and the original number.

Steps in main()

1. Read the main number.
2. Read the digit that needs to be searched.
3. Extract the last digit using ``% 10``.
4. Remove the last digit using ``/ 10``.
5. Compare the extracted digit with the target digit.
6. Increase ``freq`` when they are equal.
7. Repeat until all digits are checked.
8. Print the final frequency.

💡 Example

◆ Input:

 `Number = 122342`

 `Digit = 2`

◆ `122342 % 10 = 2` → Frequency = 1

◆ `12234 % 10 = 4` → Frequency = 1

◆ `1223 % 10 = 3` → Frequency = 1

◆ `122 % 10 = 2` → Frequency = 2

◆ `12 % 10 = 2` → Frequency = 3

◆ `1 % 10 = 1` → Frequency = 3

➡ Result

 `The digit 2 appeared 3 times in 122342.`

Note

- ◆ The same ``% 10`` and ``/ 10`` technique from the previous lessons is used to process every digit.
- ◆ ``freq`` starts at ``0`` because no occurrences have been found yet.
- ◆ The code uses ``float Digit``, but a digit should normally be stored as ``int`` because digits are whole numbers from ``0`` to ``9``.

⊘ Warning

- ◆ The program does not validate that the second input is actually between `0` and `9`. The user could enter a number such as `25`.
- ◆ Because `CountFrequency()` receives `Digit` as a `float`, the comparison still works for normal integer digit input, but `int` would be a more appropriate type.



Comparison

▨ Frequency Counter

- ✓ Starts from `0`.
- ✓ Increases by `1` every time the target digit is found.
- ✓ Example: `122342`, target `2` → `3`.

▤ Digit Extraction

- ✓ `% 10` gets one digit at a time.
- ✓ `/ 10` removes that digit.
- ✓ Example: `122342` → 2 → 4 → 3 → 2 → 2 → 1.

✦✦ Summary

- ✓ Digit frequency means counting how many times a digit appears.
- ✓ ``% 10`` extracts the last digit.
- ✓ ``/ 10`` removes the last digit.
- ✓ ``freq++`` counts every matching digit.
- ✓ The program checks all digits until the number becomes ``0``.
- ✓ ``122342`` contains the digit ``2`` three times.

Thank you 😊